

Tamweel Lite

Trainee Name: Huda Makki Mohammed

Trainer: Meaad Almarri

Academy: [SDAIA Academy](#)

Program: Advanced Machine Learning Methods

Program Code: SDA-DSC-211



Business Problem

The project aims to identify applications with a higher estimated risk of default within 90 days after application, while balancing predictive performance, simulated decision cost, and limited review capacity.



Decision Objective

Build a reliable classification workflow that assigns risk scores to applications and determines which cases should be flagged for review.



Why It Matters

Missing a potential default is treated as more costly than reviewing a non-default application. Therefore, the project uses a simulated decision cost of:

$$10 \times \text{False Negatives} + 1 \times \text{False Positives}$$



10,000

Training Applications



2,500

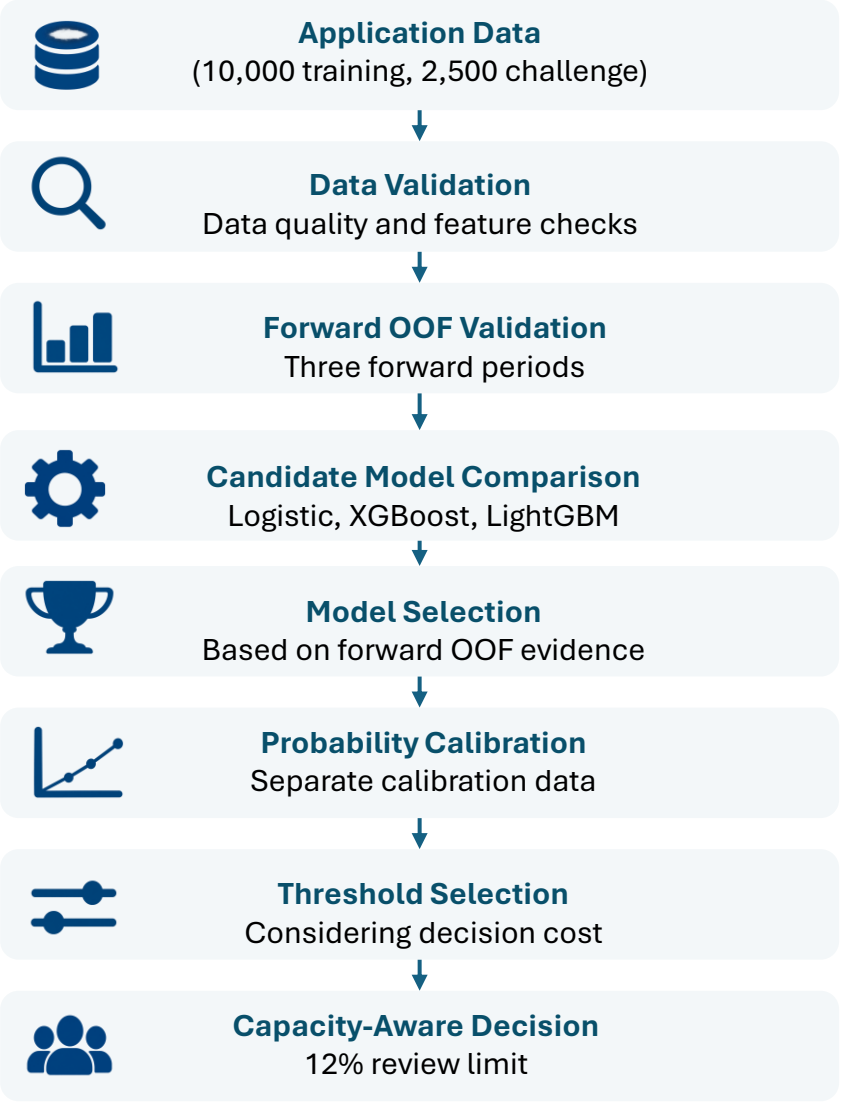
Challenge Applications



12%

Maximum Review Capacity

Leakage-Aware Modelling & Validation Strategy



Key Principles



Leakage Prevention

Temporal ordering, customer separation, and label availability were respected throughout the modelling workflow.



Forward Validation

Out-of-Fold predictions were generated using forward validation, ensuring that earlier observations were used to predict later periods.



Model Candidates

LightGBM, XGBoost, Logistic Regression, and ensemble approaches were compared using the same forward OOF evidence.



Separate Calibration

Calibration data were kept separate from model selection.



No Challenge Labels

Challenge data were not used for training or tuning.



2,155
Forward OOF
Predictions

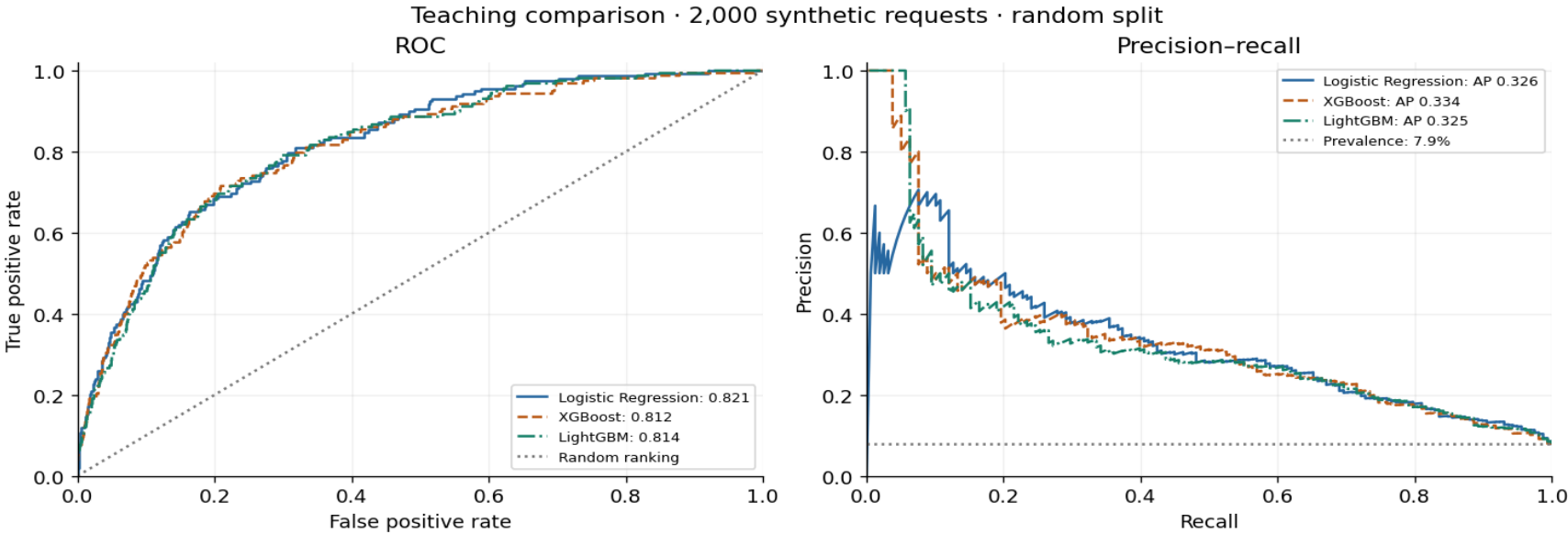


3
Forward
Validation Periods

Model Comparison — Is the Ensemble Worth Shipping?

Final Decision	Key Result
KEEP SINGLE Selected Model: Logistic Regression	Mean AP: 0.3917 Fold SD: 0.0298

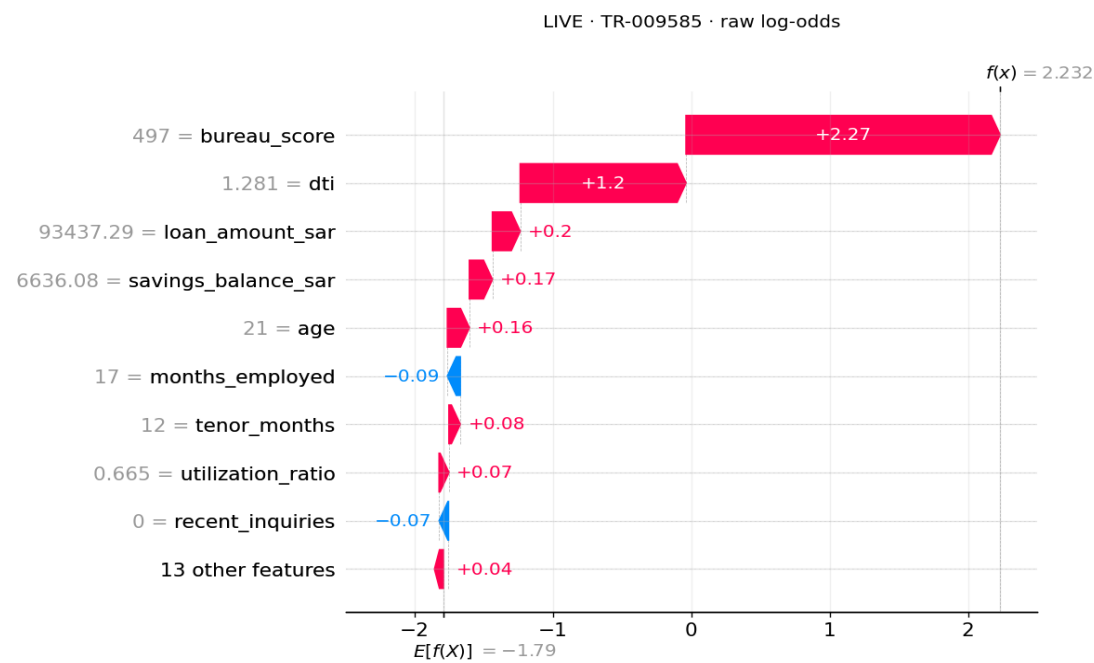
Model	LightGBM	XGBoost	Logistic	Equal Ensemble	Weighted Ensemble	Stack
Mean AP	0.3455	0.3526	0.3917	0.3717	0.3894	0.3831



Evidence Boundary The challenge set is unlabelled, therefore, no final AP, FPR, or simulated-loss performance is claimed on the challenge data.

Example High-Risk Application (TR-009585)

Raw model probability: 0.9031
Calibrated probability: 0.4795



SHAP values are expressed in raw log-odds, not probability points.

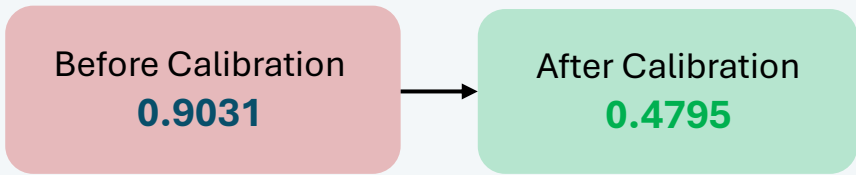
Interpretation is associative, not causal

Day 4 explanations are model-specific and should not automatically be treated as explanations of a different refitted final model.

Probability Calibration

Why Calibration Matters

A high raw model score should not automatically be interpreted as a reliable probability.



Metric	Raw	Sigmoid	Change
Brier Score	0.113027	0.067112	-0.04592
ECE	0.146871	0.022486	-0.12438

Negative changes indicate improved probability reliability.

KEEP SINGLE — LOGISTIC REGRESSION

The simpler model was selected because the ensemble did not demonstrate sufficient incremental value on forward OOF evidence.

Final Challenge Decision (2,500 applications)

